

LLM-ERP Support Runbook (English) — v3.0

What to do when things break — for customer IT or our support staff Designed so you can self-resolve 90% by following this; only the last 10% needs us.

⚡ **v3.0 Strategic Pivot Notice:** §6.3 "LINE Bot Webhook not receiving" is deprecated in v3.0 (LINE Bot is no longer a product feature).



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1. Severity Tiers

Level	Description	Response	Handler
 P0	System fully down / data corruption	Immediate	Vendor + Customer IT both
 P1	Core functions broken (cannot login / order)	Within 1 hour	Vendor
 P2	Some function abnormal but workaround exists	Within 8 hours	Vendor
 P3	UI bug / typo	Within 1 week	Vendor

2. Quick Symptom Lookup

Symptom	Try First	Details
Browser shows nothing	<code>docker compose ps</code> , see which container is down	\$3
Login then blank page	F12 → Console errors; likely JWT_SECRET changed	\$3.4
Inventory query hangs	Check backend log for N+1 query	\$4
AI chat "Connection error"	Check <code>/api/health</code> <code>llm_provider</code> status	\$5
Owner LINE Bot silent	Check Cloudflare Tunnel + LINE Channel webhook URL	\$6
"Database is locked"	SQLite write contention	\$7
Sudden 429 Too Many Requests	Rate limit triggered, possible attack	\$8

3. System Won't Start

3.1 Check container status

```
docker compose ps
```

`Exit X` → that service crashed.

3.2 Backend startup failure

```
docker compose logs backend | tail -50
```

Common errors:

Log Message	Cause	Fix
<code>JWT_SECRET still default</code>	No secret set	Edit <code>.env</code> → <code>JWT_SECRET=\$(openssl rand -hex 32)</code>
<code>OperationalError: no such table</code>	Migration not run	<code>docker compose exec backend alembic upgrade head</code>
<code>Connection refused: postgres</code>	DB not running	<code>docker compose up -d db</code>
<code>ImportError</code>	Incomplete upgrade	<code>git status</code> to find missing files

3.3 Frontend won't start

```
docker compose logs frontend | tail -30
```

Common: nginx config typo → check `nginx.conf` .

3.4 Login then blank page

90% it's "JWT_SECRET changed, old tokens invalid but browser still caches".

Fix:

1. Browser DevTools → Application → Clear site data
2. Re-login

4. API Slow / Timeout

4.1 Find slow endpoint

```
docker compose logs backend | grep "took" | sort -k 9 -nr | head -10
```

4.2 Common Causes

Cause	Evidence	Fix
N+1 query	Log shows many sequential SELECTs	Add <code>selectinload</code> in service
LLM slow	Only chat endpoint slow	\$5
Disk full	"disk full" warning	Clean old data / add storage
Missing index	Big table + slow query	<code>EXPLAIN</code> + add index

4.3 Emergency throttle

If some endpoint is being hammered:

```
# Rate limit built-in; tighten via .env
RATE_LIMIT_ENABLED=true
docker compose restart backend
```

5. LLM Unreachable

5.1 Confirm health

```
curl http://localhost:8000/api/health | jq .llm_provider
# Should return "deepseek" / "claude" etc., not null
```

5.2 Test API Key

```
docker compose exec backend python -c "  
from app.config import settings  
print('Provider:', settings.LLM_PROVIDER)  
print('API key set:', bool(settings.LLM_API_KEY))  
print('Model:', settings.LLM_MODEL)  
"
```

5.3 Provider-Specific Outages

Provider	503 Workaround
Claude	Wait 5 min / switch to DeepSeek temporarily
OpenAI	Same
DeepSeek	China DC sometimes unstable; switch to local Ollama
Ollama	Confirm Ollama service alive: <code>curl http://ollama:11434/api/tags</code>

5.4 Emergency Degrade to Demo Mode

```
# .env  
LLM_PROVIDER=disabled  
docker compose restart backend  
# All chat endpoints return demo message (no error); other functions normal
```

6. SSE Push Not Received

6.1 Check nginx config

```
docker compose exec frontend cat /etc/nginx/conf.d/default.conf | grep -A 5 "/api/events"
```

Must include:

```
proxy_buffering off;  
proxy_cache off;  
chunked_transfer_encoding off;
```

6.2 Test SSE channel

```
curl -N http://localhost:8000/api/events/stream  
# Should see heartbeat every 10 seconds
```

If silent → nginx buffering not disabled.

6.3 LINE Bot Webhook not receiving

- Cloudflare Tunnel up? `cloudflared tunnel list`
- LINE Channel webhook URL set correctly?
- Signature validation passing? Check backend log.

7. Database Locked

"Database is locked" almost always = SQLite write contention.

7.1 Immediate Fix

```
docker compose restart backend
```

7.2 Long-term Fix

Upgrade to PostgreSQL:

```
# .env  
DATABASE_URL=postgresql+asyncpg://erp:password@db:5432/erp  
# Also edit docker-compose.yml to add PostgreSQL service
```

See `docs/ADMIN_GUIDE.md` "Upgrade to PostgreSQL" section.

8. Under Attack

8.1 Detection

Run daily:

```
docker compose logs backend | grep -E "429|risk_flagged|injection" | wc -l
```

Sudden spike → possible attack.

8.2 Immediate Actions

1. Tighten rate limit (edit `.env`):

```
RATE_LIMIT_LLM_CHAT=5/minute # drop from 30 to 5
```

2. Disable LLM temporarily (if it's an LLM cost attack): `LLM_PROVIDER=disabled`
3. Add firewall rules: `deny` blacklist IP in nginx
4. Check audit log to identify source:

```
docker compose exec backend python -m scripts.audit_query \  
--period 1h --filter "status_code=429"
```

8.3 Report + Add Whitelist

Once attack source identified:

- Whitelist in nginx (only allow customer's office IP)
- Notify customer IT to rotate VPN

9. Daily Monitoring SOP

9.1 Daily (Automated)

Add to crontab:

```
# Daily 09:00 health check + LINE notify  
0 9 * * * /opt/llm-erp/scripts/daily_health_check.sh
```

`daily_health_check.sh` :

```
#!/bin/bash
cd /opt/llm-erp
HEALTH=$(curl -s http://localhost:8000/api/health)
LINE_TOKEN=$YOUR_LINE_NOTIFY_TOKEN

if [ "$(echo $HEALTH | jq -r .status)" != "ok" ]; then
    curl -X POST https://notify-api.line.me/api/notify \
        -H "Authorization: Bearer $LINE_TOKEN" \
        -d "message=⚠️ LLM-ERP health check failed: $HEALTH"
fi
```

9.2 Weekly

- Check `GET /api/analytics/ai-cost` — no overspend
- `docker compose logs backend | grep ERROR | wc -l` should be < 10
- Run `bash scripts/run_gates.sh` — 8 gates all green

9.3 Monthly

- Run backup script
- Run `scripts/data_quality_check`
- Monthly review with customer

10. Emergency Upgrade vs Rollback

10.1 Standard Upgrade

```
cd /opt/llm-erp
git fetch && git checkout v2.5 # specify version, not `main`
docker compose pull
docker compose up -d --build
# Run migrations
docker compose exec backend alembic upgrade head
# Verify
bash scripts/run_gates.sh
```


10.2 Rollback After Bad Upgrade

```
cd /opt/llm-erp
docker compose down
# Restore data
cp /opt/backup/erp-pre-upgrade.db erp.db
# Revert version
git checkout v2.4
docker compose up -d --build
```

Pre-requisite: Always backup before upgrade: `docker compose exec backend cp /app/erp.db /tmp/erp-$(date +%F).db`

If You're Stuck

Severity	Contact
P0 system down	24/7 phone (Pro+) / LINE @llmerp
P1 core broken	Business-day office hours
P2/P3	support@llm-erp.example

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